

Learning from experiences of evaluators implementing theory-driven evaluations in diverse settings: Building on the contributions of John Mayne

Sanjeev Sridharan^{a,*}, April Nakaima^b

^a University of Hawaii at Manoa, USA

^b The Evaluation Centre for Complex Health Interventions, University of Toronto, Canada

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ABSTRACT

The papers in this volume grapple with various issues related to the use of theories of change in program evaluation. This introductory paper reviews some of the critical challenges that arise in developing and learning from theory-driven evaluations. These challenges include the relationship between theories of change and ecologies of evidence, the need for epistemic fluency in learning, and coming to terms with the initial incompleteness of knowledge in program mechanisms. The nine papers that follow, which represent a geographically diverse set of evaluations and evaluators (including Scotland, India, Canada, USA), help develop these and other themes. This volume of papers also serves as a celebration of the work of John Mayne, one of the foremost theory-driven evaluators of the last few decades. John passed away in December of 2020. This volume is intended to honor his legacy while also identifying challenging issues that call for further development.

1. Introduction

We along with our co-editor (and mentor) Mel Mark work closely in the area of theory-driven evaluation. As we collaborated as a team on evaluating complex systems, we have had intense dialogues on the representational challenges of dynamic interventions and, also, how one tests and learns from key linkages and assumptions in the theory of change for dynamic systems (Van Geert, 2020). We concurred that the practice of theory-driven evaluation for dynamic systems was constrained by the artificial boundaries that one sometimes draws in the professional practice of evaluation. For example, in many evaluations we saw theorizing about how an intervention might work, with explicit discussions on assumptions and mechanisms, but with no clear focus on how such theoretical linkages were or should be empirically interrogated. In other examples, we saw many complex statistical and qualitative methods being deployed in evaluations of interventions without a clear connection to the theory of change. We sensed that the important interplay between theory and evidence (Hay, 2016; Van Maanen et al., 2007) needed far greater focus and precision.

Our discussions focused on how to more clearly develop the connection between theory and evidence. While we started from the perspective of *testing* linkages of the theory of change, we soon began to

discuss more broadly how one *learns* from theories of change. This discussion did not just focus on methodology; it raised several other high-level issues, including: how evaluation teams should be organized, thinking more broadly about what evidence means, evaluator competencies, and how learning transfers between different contexts. This volume emerged as a result of such focused discussions and debates.

Such a focus on high-level themes (with some accompanying 'lofty' questions) needed to be complemented with experiences from evaluation practitioners in applying theory-driven evaluation. As we wanted to learn from a very diverse set of experiences, we invited evaluators from Canada, India, Scotland, and the US to reflect and share issues and challenges on theory-driven evaluation that had emerged in their work. Despite our interests in larger, strategic themes in evaluation, we were not prescriptive on what the authors needed to focus on. We left it to the author's discretion on what they saw as crucial issues in learning and testing theories of change; over time as the volume proceeded, we requested a greater focus on learning about (and from) being explicit about assumptions in theories of change. We were confident that the reflections from the diverse set of authors would not just serve to connect with the strategic conceptual ideas we were interested in, but they would also generate practical 'how to' guidance on enhancing the interplay between theory and evidence. We also felt that the volume

* Corresponding author.

E-mail address: sanjeev@hawaii.edu (S. Sridharan).

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would be enhanced with a concluding paper by our co-editor Mel Mark that summarizes some ways forward, to be more tactical (and concrete) in the interplay between theory and methods.

We felt that this dialogue could benefit with newer voices who work in diverse contexts also joining this discussion on the interplay of theory of evidence. In our experience working in North American settings, often such dialogues are confined to a few elite academics and consultants. There was value in learning from voices who were new to evaluation. There was also value in learning from voices who were unconvinced that theory-driven evaluation has the answers to key issues that confront us in social decision-making for the future.

We fully expected each of the papers to take only modest steps towards highlighting some of the challenging themes that inspired this volume; progress in science proceeds with such modest steps and is often driven by the pragmatism that even small progress on difficult questions—some of the reflective variety only serves to raise the salience of difficult questions—can lead to progress in a field. Progress in addressing lofty questions will often be necessarily modest, but needs to be driven both by an opportunistic view of learning that is open to the experiences of evaluation practitioners working in diverse settings, and a recognition that the science of evaluation (Pawson, 2013) will be developed through a number of small insights (perhaps complemented with a few breakthrough insights).

1.1. Remembering John Mayne

We were joined in this endeavor by our dear friend John Mayne. This volume is inspired by our friendship and engagement with John. He was a true champion of theory-driven evaluation, believing passionately in its power, not only to help guide social and policy decision-making, but also to contribute to a just and equitable society. John was the developer of one of the more widely used theory-driven evaluation approaches called Contribution Analysis. He also contributed immensely in a number of areas including learning from evaluations, building an evaluation culture in organizations, and was particularly interested in practice related to developing ‘useful’ theories of change. We worked closely with John over the last decade on projects in Canada and India. In every one of our collaborations, he pushed us to think in more rigorous and precise ways about the nature of the interventions and about the utility of theory-driven evaluation to help inform the evaluation of the interventions. His call for rigor was accompanied by an equal measure of warmth and kindness as he connected with policy-makers, often by speaking plainly to their concerns. John inspired us to set up the Evaluation Centre for Complex Health Interventions (torontoevaluation.ca) and generously contributed to building capacities for evaluating complex interventions by being a guest of the Centre on multiple occasions. We will remember our intellectual and personal journeys with John with great fondness. John passed away in December, 2020. This volume is dedicated to his memory and the absence we feel as we miss him both as a friend and collaborator. John actively engaged with us in the production of this volume and especially in discussions of the strategic challenges that the field of evaluation faces.

1.2. Organization of the introduction

This introduction is divided into two parts. First, we discuss the eight themes that emerged in our multiple dialogues with John and Mel, framing these themes as questions that we hope will spur greater scholarship in applications of theories of change in evaluation. The themes provide a wish list of issues that we feel are important in the field of theory-driven evaluation. These themes build on a prior collaboration, under John’s leadership, on a special volume on Contribution Analysis (Mayne, 2012; Sridharan & Nakaima, 2012).

As noted above, our approach to this volume was to invite practitioners of evaluation from India, the US, Canada, and Scotland to share their reflections and experiences in implementing theories of change,

especially how they have dealt with the assumptions within a theory of change. Given our focus on not being prescriptive, we did not share the strategic themes with the invited authors. The resulting papers connect to the eight themes that emerged during our interactions and work with John and Mel. In the second part, we briefly describe each of the papers and how they contribute to some of the themes. We recognize that, given the wide scope of the themes, most of the contributions would only begin the process of explicating some of the challenging issues associated with applying theories of change.

2. Strategic issues facing theories of change

Fig. 1 below names the eight strategic themes that emerged in our discussion and the relationships between the themes.

2.1. Ecologies of evidence

As described in our 2012 paper (Sridharan & Nakaima, 2012), an evaluation has the potential of providing multiple different kinds of evidence, including evidence on:

- organizational structures and processes needed,
- program mechanisms,
- average impacts,
- differential impacts for different segments of the population,
- what is needed to sustain the program,
- implementation needs over time,
- connections between the program and other programs in the community,
- program contexts and support conditions that matter for facilitating program impacts.

The above are just a few examples of the types of evidence that evaluations can generate, though often does not. A key focus of our 2012 paper was on the patterns of relationships between these different types of evidence. We felt that evaluation as a field could be enhanced by adopting a more reflective stance on the nature of evidence itself and on the relationships among different kinds of evidence. Evaluators sometimes speak about evidence in a narrow sense – for example, evidence of the average impact of an intervention. In our view, theory-driven



Fig. 1. Volume Themes.

evaluators need to be more explicit about the relationships between theories of change and the varieties of evidence that can be generated by evaluations.

As Cartwright et al. (2009, p. 5) argue, there is a need for greater clarity on both the concept of evidence and how best to combine evidence in testing hypotheses:

“... we do not have a theory of evidence that can be called upon in policy deliberations. We are supposed to base our policies on evidence but how exactly are we to proceed: what is to count as evidence and how shall we use it? ... we lack a practicable theory of evidence – one that can be put to use for evidence-based policy. There are three essential ingredients missing. We do not have:

- A reasonable and practicable concept of evidence
- A reasonable and practicable account of what different pieces of evidence say about a hypothesis and with what strength they speak...
- A reasonable and practicable account of how to evaluate a hypothesis in the light of all the candidate evidence.”

This focus on a general theory of evidence for policy-making is highly relevant to this volume. Such a theory of evidence could help build coherence across the different kinds of evidence that can be generated from an evaluation or multiple evaluations.

2.2. The need for epistemic fluency

Working in multiple settings in Canada and India, and in multiple indigenous settings in Hawaii, India, and Canada, we are acutely aware of the cultural dimensions of knowledge and the need to respect different ways of knowing. We find the concept of ‘epistemic fluency’ useful in helping set the stage for the kinds of knowledge that are needed given that different communities have heterogeneous views about what knowledge matters and also how best to learn about how programs work. Markauskaite and Goodyear (2017, p.1) define *epistemic fluency* as “working on real-world problems usually requires the combination of different kinds of specialised and context-dependent knowledge, as well as different ways of knowing. People who are flexible and adept with respect to different ways of knowing about the world can be said to possess epistemic fluency.” Epistemic fluency points to a gap we have seen in the field – a lack of interest among many evaluators in exploring epistemological questions around how we learn about the world and how best to combine different kinds of knowledge. Our work in evaluation capacity building in multiple settings (Sridharan et al., 2017) highlights the need to question our certainties about how different cultures might view assumptions in a theory of change. For the most part, theory-driven evaluation has not seriously explored the cultural dimensions of knowledge or, more specific to our interest in this volume, the cultural dimensions of assumptions in theories of change. It is important to ask, for example: in what ways does reflecting on a theory of change, including a specific assumption within the theory of change, lead to a recognition of the cultural contexts of the assumption?

Stimulated by their collaborative work in India, John, Mel and Sanjeev discussed the mechanical nature of how theory-driven evaluation is often practiced. We were disappointed by the many examples of theories of change that were developed without recognition (or even a minimal understanding) of the specific cultural, community, or policy contexts in which an intervention was embedded. A focus on epistemic fluency would emphasize contextual considerations, and also help highlight what competencies matter for evaluators and for evaluation training. For example, in our discussions on what constitutes a good theory of change, surprisingly there has been limited discussion on the importance of capturing the diversity of stakeholder voices in the development of the theory of change. How do we ensure that we are not imposing our epistemologies over a community’s way of knowing? But

even this framing is somewhat limited, because even within communities there are diverse voices and cultures. How do we tell theory-driven evaluation stories that are respectful of the diversity *within* communities as well *between* communities? Given the debates within the evaluation field about qualitative and quantitative approaches, for example, relatively few evaluations explicitly combine different methodological approaches in testing assumptions in theories of change (leave aside different ways of knowing). Much work remains to be done.

2.3. The need for a focus on the incompleteness of knowledge

Many of our evaluations have been informed by a realist perspective, which recognizes that programs are complex systems thrust into other complex systems (Pawson et al., 2004; Sridharan & Nakaima, 2011). Yet, despite acknowledging the complexities of the world in which programs are implemented, we still often pretend to know more than we do about how programs actually bring about change. In our observation, very few theories of change, even those grounded in realist evaluation, have been explicit about representing uncertainties or the incompleteness of knowledge in the initial theories of change.

Having a visual representation of a program’s theory of change is not the same as having a deep understanding of the different ways in which a program might work in different contexts. In our view, the field needs to be far more explicit about areas in which knowledge of how programs work may be incomplete. We have seen very few discussions around the lack of clarity in the assumptions themselves. Our discussions with John explored how thinking about assumptions in the theory of change could help reveal uncertainties in key areas of knowledge about how programs might work. The theory of change approach provides one way of building knowledge over time that can help us be more explicit about the incompleteness of knowledge. In our own experience, both evaluating from a distance as well as working within communities, one area of uncertainty is a lack of understanding on what aspects of the local context matter and how such contexts might shape the success of the program.

2.4. Learning about multiple contexts

Related to the preceding point is the need to be more explicit about the multiple roles of contexts and the relationships of these contexts in influencing the linkages in a theory of change. In our work, we have sought to frame contexts in terms of what we call the five-I’s: infrastructural, institutional, interpersonal, individual, and intersectional. The first four-I’s have been described in multiple realist publications by Ray Pawson and colleagues (e.g., Pawson et al., 2004). Our focus on the fifth “I” – intersectional – was especially important in thinking about intersectionality in multiple contexts in evaluations of health inequities. Working both as practicing evaluators and funders, we have been surprised that, in most applications of theories of change, descriptions of how programs work often fail to understand and incorporate *the multiple dimensions* of context. Even in those theories of change that do represent context, most that we have seen have been mechanical and superficial.

According to Vogel (2012, p. 3), there is a need for both a technical and a reflective approach to incorporate an understanding of contexts in developing theories of change: “The mapping of the logical sequence is strengthened by critical thinking about the contextual conditions that influence the programme, the motivations and contributions of stakeholders and other actors, and the different interpretations (assumptions) about how and why that sequence of change might come about.”

Thinking about contexts also raises questions about the heterogeneities in the mechanisms by which programs work differently for different groups of individuals. Realist evaluation, with its focus on contexts, is promising. Still, its focus on context needs to be more explicitly considered in thinking about program assumptions. For example, would assumptions be different for different segments (contexts) of the population? In our work, we have tried to think of iterative

evolutionary processes to incorporate contexts into developing theories of change (Sridharan et al., 2016): “In our experience, the actual practice of evaluation tends to be siloed: program developers, methodologists and program implementers rarely discuss how learnings about heterogeneities can be utilized to build the program theory as well as to improve the program over time” (p. 94).

Also, we make the case for learning cycles between theory, data analysis, and pattern recognition in developing the theory of change over time (Sridharan et al., 2016, p. 97). In the last year of John’s life, questions on contexts and theories of change preoccupied our discussions. For example: In an evaluation or implemented theory of change, how does one know if the evaluators have an understanding of the contexts? Can such knowledge be developed iteratively? Why is a good understanding of contexts not a measure of the rigor of an evaluation? Can iterative cycles of theorizing, data collection and analysis, pattern matching and recognition help surface the role of contexts in developing theories of change?

2.5. Who is learning and how is learning occurring?

In our 2012 paper (Sridharan & Nakaima, p. 389), we framed the learning problem as follows:

“Evaluations by themselves do not generate learning. A structure, process and mechanisms need to be developed to ensure the information from an evaluation actually generates learning. ...What are experiences of learning structures that have led to greater learning from evaluations? What role did different stakeholders play in such a structure?”

In our experience, limited attention has been paid to the explicit features of the structures and processes that facilitate learning. John has published extensively on building a learning culture and thinking intentionally about evaluations and learning (Mayne, 2017; Mayne, 2020). Although John has been a champion in promoting a learning culture, we feel that a theory of learning needs to be connected more critically to the theory of change of the intervention itself. There needs to be greater clarity in the field about how reflections of key assumptions promote learning. How do the local features of a context fit into a learning framework?

Reflecting on learning also raises questions about the role of an evaluator. Is the role of an evaluator to be the distant objective voice, or does the evaluator have to be deeply immersed in the context of a program to promote learning? Does the application of a method by itself generate evidence that has learning implications, or does learning require acts of co-construction with stakeholders?

2.6. Spaces for dialogue and co-construction

Evaluation practice commonly takes a deliverables/knowledge products stance. Evaluators sometimes stop after applying a certain method, sharing the findings from applications of specific methods and approaches, without bothering to involve key program stakeholders and clients in meaning-making or an interpretive process around learnings about the theories of change. In our experience, even the commissioners of an evaluation rarely create spaces for dialogue for the co-construction of meaning and the generation of recommendations. In our experience, a culture of deliverables has often trumped a culture of learning in both the commissioning and doing of evaluations.

The construction of spaces for dialogue has been prominent in our minds as we navigate the ‘product-process’ tensions in evaluation. As described by Auspos et al. (2014, p. 76):

“Change is a process, and like any process it takes time. Some practitioners believe that the process of being inclusionary, of developing a collective vision and plan for action, and of building trust, must precede action—however long it takes. Others feel

pressured to produce tangible results more quickly. The community change field calls this the process-product tension.... [T]here are no shortcuts to being inclusionary and building social capital, but there are practices that can help groups navigate the process better... [such as using] a planning and implementation process that matches the reality of the empowering and self-organizing processes.”

Paying attention to spaces for dialogue becomes especially important given the power relations that often exist in implementation and evaluation settings. Vogel (2012, p. 59) noted the relevance of “all the process principles that support a good theory of change process ... [including] dialogue with stakeholders, acknowledging multiple viewpoints and recognition of power relations, political, social and environmental realities.”

As noted previously, testing of and learning from assumptions require more than data – they also may require deep knowledge of contexts. If the theory-driven evaluation approach is to flourish, we believe the evaluator cannot be privileged to make conclusions and recommendations without involving key program stakeholders who might know more about the program contexts. Such deep knowledge of the dynamic program contexts may be critical in navigating towards outcomes. As such, *spaces (and processes) of dialogue might be needed*. Further, collaboration between the program and the evaluator should not be limited to data collection or the theorizing process; it should also include the meaning-making/interpretation of the process, which often involves the interplay of theory and evidence and a deep understanding of contexts. What kinds of processes help build dialogue and understanding of context? We are particularly interested in specific examples of the types of collaborative spaces that have led to meaningful learning from theories of change.

2.7. Timelines and the dynamics of change

Simple connecting arrows that link program activities to outputs and outcomes can often understate and trivialize the complexities of the change process. One aspect of this involves the timeline of change (Sridharan, Campbell & Zinzow, 2006) and the dynamics involved. Depending on the theory of change, a single arrow can represent a change process that might take anywhere from a few seconds to multiple decades (and sometimes longer), depending on the intervention and the context of implementation. This problem is not only related to the timeline of impact, but also has implications for the anticipated trajectory of change. Woolcock (2009) argues persuasively about the importance of considering the anticipated trajectory of change in evaluation frameworks and regarding the need to consider non-linearities and non-monotonic processes as part of understanding social changes.

John Mayne was fond of saying that, despite the apparent linear structure of some theories of change, the implied change process could still imply a non-linear process (through the presence of multiple feedback loops). Despite this insight, we feel far too much of the field still tends to uncritically accept an implicit linear view of the change process. In our experience, very few theories of change take the time to discuss the anticipated temporal features of the change process. As Woolcock (2009, p. 7) argues, the field must do better. Referring to conditional cash transfer interventions, Woolcock notes that their implicit theory of change assumes a linear process and argues instead that “a formal and explicit articulation of a theory of impact trajectory is needed nonetheless if a correct verdict on impact is to be rendered.”

Woolcock further argues that discerning such impact trajectories is both project and context-dependent and is an open question in evaluation (2009, pp. 6–7): “generating a defensible sense of the underlying impact trajectory of the project intervention...is most important...for participatory development projects, precisely because the nature of their impact trajectory is most likely to be different from other (more orthodox) projects, and indeed from each other. As argued above, the accumulated wisdom of social theory over two hundred years strongly posits

that institutional change is wrought with conflict, and often proceeds fitfully: hard won gains may endure for years, only to be eroded or eliminated altogether. Punctuated equilibriums and ‘J’ curves are the likely shape of project trajectories seeking to empower marginalised groups, yet exactly how long the periods of stasis or how deep and enduring the bottom of the curve will be as varied as the idiosyncrasies of the context in which they take place...Careful and extended engagement with each context, not hopeful extrapolation on the basis of results obtained elsewhere, should be the basis on which such a local theory is crafted and, in turn, subsequent evaluation decisions are based.”

2.8. Connecting theories of change to planning for sustainability

In recent years, there has been a greater focus on thinking about not just impacts, but also on whether such impacts are sustainable over time. In our work, we suggest that theories of change that explicitly accommodate planning for sustainability include thinking that can be different from theories of change that focus on immediate impacts (Sridharan & Nakaima, 2019). Our view is that, even in theorizing about the change process as it relates to a specific intervention, there is a need to think in an a priori way about how such change can be sustained after the funding for the intervention ends. As explained more fully below, theories of change need to more explicitly incorporate planning for sustainability (Sridharan & Nakaima, 2019, p. 378): “Decisions about sustainability often need to be made well before the evaluation provides evidence of impacts—this is because, as noted above, realistic timelines of impacts are often not considered in planning evaluations (Cook, 2000) and because decision-making cycles by policymakers may not be aligned with the timing of evaluation results (Leviton & Hughes, 1981). Hence, planning for sustainability needs to happen much earlier, and we argue that planning for sustainability should be an integral part of what we consider to be a useful theory of change. This is important because incorporating planning for sustainability can change the nature of the program itself (Sridharan & Gillespie, 2004).”

In his last few years, John had begun to incorporate a focus on sustainability in thinking about theory of change models: “[...] as Sridharan and Nakaima (2019) point out, planning for sustainability in interventions is rarely done, and in developing ToCs for interventions, identifying sustainability concerns is rarely undertaken. More often, toward the end of an intervention, questions may be asked about sustainability, perhaps to be addressed in an evaluation” (Mayne, 2020, p. 204). Further, Mayne had also started to see sustainability as critical to the theory of change *development* process. He wrote about, identifying actions that may be needed to support sustainability, including possible feedback loops, extending theories of change in ways that would “either identify risks to the sustainability of the benefits or—the preferred case—identify actions that need to be added to the design and implementation of the intervention” (Mayne, 2020, p. 219).

Evaluation as a field needs to make this focus on sustainable impacts more central. For example, given the salience of the Sustainable Development Goals, it becomes increasingly important to ask how programs can be institutionalized/routinized into existing systems as one avenue to sustain changes over time.

3. Brief summary of the papers

The above points occupied many years of our discussions with John. While none of us pretended to have answers to the challenges raised above, we sought to address them by creating spaces for dialogue with communities of evaluators and implementors. To take the dialogue forward, we often used the platform afforded to us by webinars organized by the Evaluation Centre for Complex Health Interventions (torontoevaluation.ca). John strongly believed that these types of themes could be further developed as we learn from evaluators who apply theory-driven evaluation in different settings.

While we adopted a broad canvas of reflections on theories of change, we tried to focus this volume on learning and testing of *assumptions* in a theory of change. As described by Nkwake and Morrow (2016), “assumptions are generally understood as beliefs, expectations or considerations that are taken for granted about how the world works.” Nkwake and Morrow discuss the different categories of assumptions that are possible including: ontological (“concerns the nature of the evaluand”), epistemological (“ways in which evaluand features can be ascertained”), and axiological (“related with stakeholder values”). Nkwake (2013) discusses many assumptions related to the design of programs, including: normative, diagnostic, prescriptive, causal, and external. Mayne (2015) also identifies different types of causal assumptions, including: reach, capacity, behavioral, and wellbeing.

Fig. 2 summarizes the connections between each of the papers in this volume and the eight themes discussed above.

The first paper in this volume is by John Mayne himself. Responding to the lack of an explicit definition of what constitutes assumptions, John finds it useful to differentiate between what he calls the *cause-effect assumptions* and the *causal-linkage assumptions*. Cause-effect assumptions serve to explain the relationship between a cause and effect, while causal-link assumptions describe the events and conditions necessary for the causal-linkage to work. John also discusses the COM-B model, which focuses on the roles of capabilities (C), opportunities (O), and motivations (M) in explaining behavior (B). The paper connects John’s framework of theories of change with realist evaluation (Pawson & Tilley, 1997). Mayne ends with an appeal to be more precise regarding assumptions in a theory of change. The paper holds added significance because it is one of John’s final papers, completed a few weeks before he passed away.

The remainder of the papers can be grouped roughly into three categories. First, we have papers that engage with the initial incompleteness of theories, with both Mark and Nakaima et al. arguing for data-driven approaches to learn about the theory of change. We then move on to papers that think deeply about the lived experiences of clients and how theories of change can best capture such realities. Cook et al. focus on the significance of relationships, Hazra et al. address the importance of dialogue spaces, or more humbly understanding the context of client lives, and Sridharan et al. suggest the need to start with clients’ lived experiences and be more sensitive to the heterogeneous realities of different kinds of clients. The two final papers in the volume discuss the challenges of learning about systems and making judgments about the future. Das and Singh explore how John’s work might be extended to system level thinking, whereas Carden suggests that radical humility is needed to make judgments about the future based on learnings about the past.

Mel Mark’s paper describes the role of analysis in helping unpack incomplete theories change. Mark highlights the importance of “elliptical assumptions” that “are not explicit in the theory of change ...” (Mark, this volume) to better understand initially unknown but important ingredients for programs to be effective. He argues that paying attention to data, analysis, and pattern recognition incorporating a focus on outliers and differential patterns of findings will help identify learnings about new mechanisms that the original theory of change had not considered. The examples that Mark provides for elliptical assumptions come from the field of education: one on a pre-algebra curriculum and a second on class size. They illustrate the need for an interplay between theory and evidence to more completely develop and refine theories of change. Mark recognizes that such assumptions are ubiquitous: “Our knowledge is sufficiently limited that we should usually assume that elliptical assumptions exist, that is, that there are important ingredients for program success which are not yet explicit in our theories of change.” Mark’s paper raises important questions: Given the ubiquity of such assumptions, who is responsible for surfacing such elliptical assumptions? What kinds of collaborative spaces facilitate the surfacing of such assumptions?

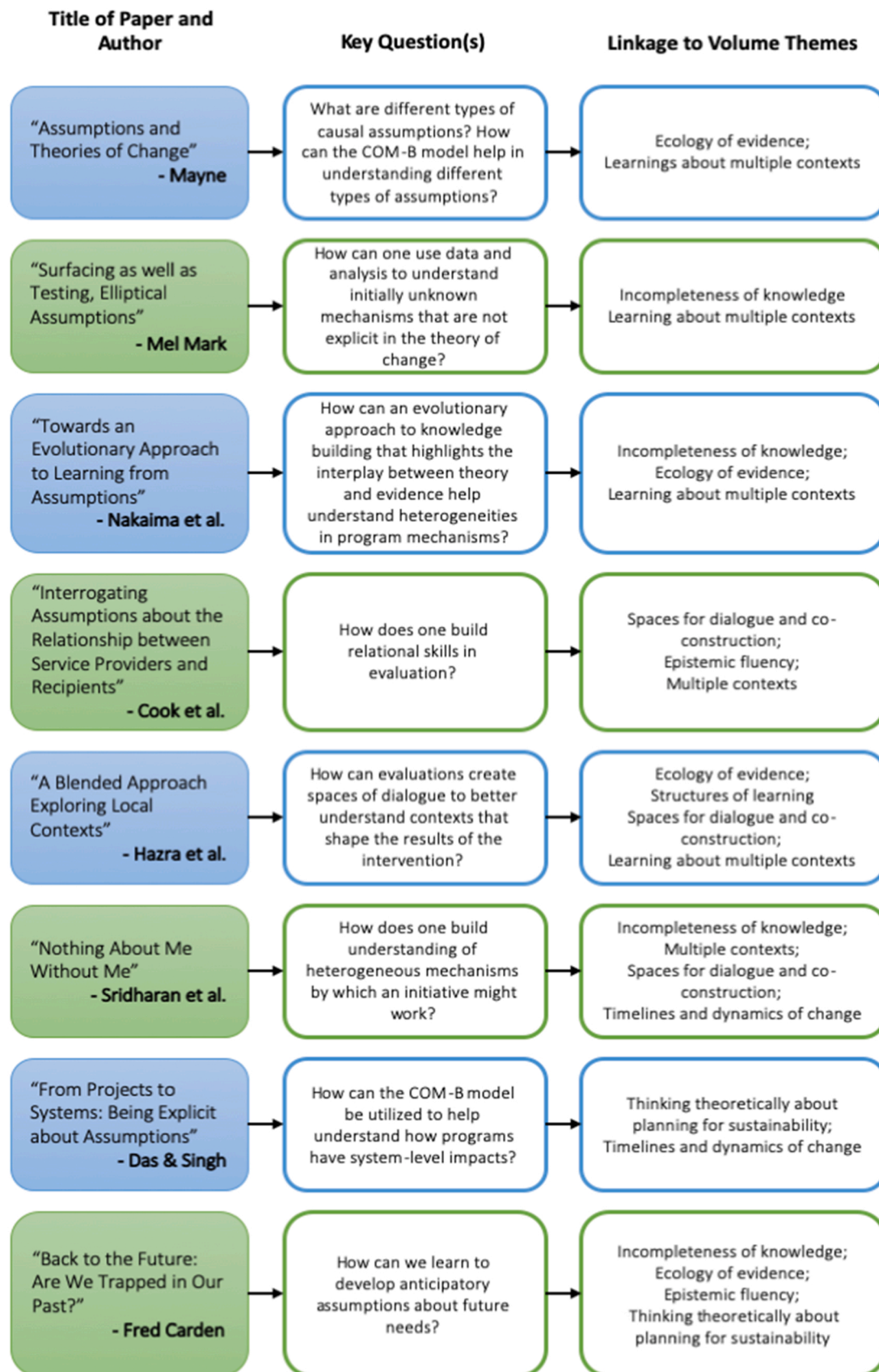


Fig. 2. Summary of Papers in the Volume.

Nakaima et al. further build on the insight that our knowledge is often incomplete at the outset of developing theories of change. They argue for an evolutionary approach to the interplay between theory and evidence to better understand heterogeneities in mechanisms when faced with knowledge incompleteness in the theory of change. Applying this evolutionary approach to a Toronto-based dance program designed for individuals with Parkinson's disease (PD), Nakaima et al. demonstrate that the theory of change of how dance could make a difference in the lives of Parkinson's clients was quite unclear at the outset. Given that PD is a neurodegenerative disease, it is likely that different heterogeneous sets of interventions ('causal bundles') might be needed for different individuals. The authors argue that trying to understand 'what works for whom' in such contexts might require a dynamic, evolutionary approach to knowledge building: "Taking an evolutionary approach suggests evaluation purposes beyond summative in helping build knowledge of heterogeneous causal bundles. This might include planning for a portfolio of evaluations with various purposes and designs to answer sets of questions and to progressively incorporate and verify knowledge as it is generated. We need to learn from assumptions and better understand the contexts under which the assumptions might hold" (Nakaima et al., this issue).

Cook et al. suggest there is a need to be more explicit about the importance of relationships in understanding how programs work. Cook et al. persuasively argue for the need for evaluators to take a relational approach to interrogating assumptions: "Whilst there are many examples of evaluation practitioners focusing on relationships between providers and recipients of service, theory of change models have tended to give primacy to cognitive rather than relational components of a change process" (Cook et al., this issue). They further argue that "There is a risk that the importance of interpersonal relationships is taken for granted or that trusting relationships are assumed. Including relationships explicitly within a theory of change is an important way to avoid this risk" (Cook et al., this volume). Given the importance of interpersonal competencies in evaluation (King et al., 2001; King & Stevahn, 2018) and the importance of relationships in most social change processes, we believe relationships between key stakeholders involved in the change process need to be more central in our theories of change. A critical question for the field is: How does one build such relational skills in evaluators? In our experience, interpersonal competencies are often not given the same attention as technical competencies.

Hazra et al. discuss an intriguing example from a gender empowerment intervention by focusing on self-help groups in Uttar Pradesh, India. They look at whether membership in such self-help groups can help improve maternal, newborn, and child health practices. Their paper reflects on how multiple methodological approaches are needed to leverage learning. Theory-driven evaluators should ideally incorporate a process and spaces of dialogue for co-construction of meaning to better understand the context that might shape the results of an intervention. Hazra et al. persuasively demonstrate the importance of context when learning about assumptions, the critical need for mixed methods in learning, and the need to prioritize the learning needs of clients and other stakeholders: "the needs of a funder or an academic/consultancy organization are often implicitly considered higher than the needs of the program participants. In the co-construction dialogues, it became obvious that community members themselves appreciated the dialogues, and they had far deeper insights into the weaknesses of our statistical models including key gaps in our theorizing and measurement. Such knowledge only surfaced after we created spaces in which their views were treated not only as validation of the results but as a constant process of refinement and development in understanding the assumptions in a theory of change" (Hazra et al., this issue).

Sridharan et al. describe the central importance of starting with the clients themselves in the development of the theory of change. Their focus is on drop-in/shelter services for homeless clients in the city of Toronto. The authors suggest that it is highly problematic to develop initial theories of change that do not involve clients while using client

data only to refine or validate a theory of change. In their paper titled, "Nothing About Me Without Me," Sridharan et al. argue that the theories of change developed through talking to funders and other program staff were quite different from those developed by talking to clients. The authors stress that theories of change must be especially sensitive to the needs of marginalized individuals and also pay attention to learning about the dynamics of marginalization: "... when working in settings that involve marginalized individuals and the evaluation/research team's lived experience is limited, it is important that we do more than simply 'engage' clients; we must be almost entirely led by clients' perspectives in developing the initial theory of change" (Sridharan et al., this issue).

Das and Singh focus on a system-level intervention of a human resource management system implemented in the Indian state of Uttar Pradesh. They build on Mayne's theory of change model to explore how his applications at the individual level can also be extended to systems-level work. Das and Singh demonstrate how evaluation questions can flow from initial theories of change to help sharpen the evaluation's focus on systems-level processes. Their paper is a reflection on how Mayne's thinking can be extrapolated to the systems level to describe how programs can have system-level impacts. One of the implications of their paper is the need for theories of change to not only understand how interventions work, but also to more clearly describe how systems can absorb and sustain innovations over time: "The paper also highlights the need for theories of change to more clearly conceptualize and define the construct of how systems can absorb and maintain innovations — the absorptive capacity of systems — by this we simply mean the ability of systems to absorb, institutionalize and maintain the programmatic innovations over time" (Das & Singh, this issue).

Fred Carden, who worked closely with John in both Canada and India, takes a more skeptical view on the utility of learning from assumptions in theories of change for making judgments about the future. He argues that making statements about the future requires us to be less than comfortable in relying on the results of any testing or learning process from the theory-driven evaluation. Looking at the relationship between the past and future, Carden argues that the focus needs to be not on the program assumptions of past interventions, but on the anticipatory assumptions of future needs: "Exploring anticipatory assumptions is crucial, as is freeing ourselves from the assumptions we hold that the future will mirror the past. These limit our view of the future to what we have learned from the past, colonizing our thinking to the maintenance of the status quo" (Carden, this volume). Carden's paper also underscores the importance of being humble about the cultural contexts of an assumption: "This is further complicated when we are working across cultural settings when our own assumptions may be quite different from the contexts and ways of knowing in which we are working..." We would like to note that John would have wholeheartedly welcomed Carden's skepticism as an important pre-condition for the advancement of the science of evaluation (Pawson, 2013). We also believe that Carden's paper has important implications for how evaluators are trained, as we learn that our recommendations from theory-driven evaluation are not just for the immediate future, but can also be for the long-term future. And, while Carden does not make thinking and planning for sustainability a central feature of his paper, there are important lessons in his paper on how we train evaluators to think of longer-term sustainability.

The concluding paper is by Mel Mark. Mel has long been a champion of theory-driven evaluation, and in his work, he has sought to be concrete about the interplay between theory, data, and analysis. As his paper on elliptical assumptions in this volume also reflects, he has not been hesitant to use data-driven approaches to more fully explicate program mechanisms. In the final concluding piece titled, "Doing Better in Theory-based Evaluation, from A to Z," Mark reflects on what the field needs, and offers tactical and concrete guidance to strengthen the interplay of theory and evidence.. Mel worked closely with John on health systems work in India and contributed to shaping many of the

themes discussed in this volume.

4. Conclusions

John Mayne passed away before this Special Issue was completed. He would have welcomed such an international focus on different aspects of applying theories of change, but would have been uncomfortable with a focused gaze on his work (and his legacy). He was a quiet person who liked dialogue, building community, and engaging in debate in respectful and constructive ways. He thrived on differences of perspectives. As we attempt to help build the field, we return to Don Campbell's original vision of building a "mutually monitoring, disputatious community of scholars" (Campbell, 1984, p. 44) who can learn and build on forums such as the papers represented in this volume.

References

- Auspos, P., Caba, M. The Aspen Institute. (2014). Complexity and Community Change: Managing Adaptively to Improve Effectiveness. Retrieved September 29, 2022, from https://assets.aspeninstitute.org/content/uploads/files/content/docs/pubs/Complexity_and_Community_Change.pdf.
- Campbell, D. T. (1984). Can we be scientific in applied social science? In R. F. Connor, D. G. Altman, & C. Jackson (Eds.), *Evaluation studies review annual* (Vol. 9, pp. 26–48). Newbury Park, CA: Sage.
- Cartwright, N., Goldfinch, A., & Howick, J. (2009). Evidence-based policy: Where is our theory of evidence? *Journal of Children's Services*, 4(4), 6–14. <https://doi.org/10.5042/jcs.2010.0017>
- Hay, M. C. (2016). *Methods that matter integrating mixed methods for more effective social science research*. Chicago: The University of Chicago Press.
- King, J. A., Stevahn, L., Ghore, G., & Minnema, J. (2001). Toward a taxonomy of essential evaluator competencies. *American Journal of Evaluation*, 22(2), 229–247. [https://doi.org/10.1016/S1098-2140\(01\)00129-1](https://doi.org/10.1016/S1098-2140(01)00129-1)
- King, J. A., & Stevahn, L. (2020). Presenting the 2018 AEA evaluator competencies. *New Directions for Evaluation*, 2020(168), 49–61. <https://doi.org/10.1002/ev.20435>
- Markauskaite, L., & Goodyear, P. (2017). Epistemic fluency and professional education. *Professional and Practice-based Learning*. doi:10.1007/978-94-007-4369-4.
- Mayne, J. (2012). Contribution analysis: Coming of age? *Evaluation*, 18(3), 270–280. <https://doi.org/10.1177/1356389012451663>.
- Mayne, J. (2015). Useful theory of change models. *Canadian Journal of Program Evaluation*, 30(2), 119–142. <https://doi.org/10.3138/cjpe.230>
- Mayne, J. (2017). Structuring evaluations and monitoring for learning. *Success in Evaluation: Focusing on the Positives*, 22, 47–68. <https://doi.org/10.4324/9781315130422>
- Mayne, J. (2020). Sustainability analysis of intervention benefits: A theory of change approach. *Canadian Journal of Program Evaluation*, 35(2). <https://doi.org/10.3138/cjpe.70004>
- Nkwake, A. M., & Morrow, N. (2016). Clarifying concepts and categories of assumptions for use in evaluation. *Evaluation and Program Planning*, 59, 97–101. <https://doi.org/10.1016/j.evalprogplan.2016.05.014>
- Pawson, R., & Tilley, N. (1997). *Realistic Evaluation*. London: Sage.
- Pawson, R., Greenhalgh, T., Harvey, G., & Walshe, K. (2004). Realist Synthesis: An Introduction. ESRC Research Methods Programme, University of Manchester. RMP Methods Paper 2. Retrieved September 29, 2022, from <http://www.ccsr.ac.uk/methods/publications/RMPmethods2.pdf>.
- Pawson, R. (2013). *The science of evaluation: A realist manifesto*. SAGE Publications Ltd. <https://doi.org/10.4135/9781473913820>
- Sridharan, S., & Gillespie, D. (2004). Sustaining problem-solving capacity in collaborative networks. *Criminology & Public Policy*, 3(2), 221–250. <https://doi.org/10.1111/j.1745-9133.2004.tb00038.x>
- Sridharan, S., Campbell, B., & Zinzow, H. (2006). Developing a stakeholder-driven anticipated timeline of impact for evaluation of Social Programs. *American Journal of Evaluation*, 27(2), 148–162. <https://doi.org/10.1177/1098214006287990>
- Sridharan, S., & Nakaima, A. (2011). Ten steps to making evaluation matter. *Evaluation and Program Planning*, 34(2), 135–146. <https://doi.org/10.1016/j.evalprogplan.2010.09.003>
- Sridharan, S., & Nakaima, A. (2012). Towards an evidence base of theory-driven evaluations: Some questions for proponents of theory-driven evaluation. *Evaluation*, 18(3), 378–395. <https://doi.org/10.1177/1356389012453289>
- Sridharan, S., Jones, B., Caudill, B., & Nakaima, A. (2016). Steps towards incorporating heterogeneities into program theory: A case study of a data-driven approach. *Evaluation and Program Planning*, 58, 88–97. <https://doi.org/10.1016/j.evalprogplan.2016.05.002>
- notes. In Sridharan, S., Zhao, K., & Nakaima, A. (Eds.), *New Directions for Evaluation*, 2017, (pp. 9–16). (2017) pp. 9–16. <https://doi.org/10.1002/ev.20242>
- Sridharan, S., & Nakaima, A. (2019). Till time (and poor planning) do us part: Programs as dynamic systems—incorporating planning of sustainability into theories of change. *Canadian Journal of Program Evaluation*, 33(3), 375–394. <https://doi.org/10.3138/cjpe.53055>
- Van Geert, P. L. C. (2020). Dynamic systems, process and development. *Human Development*, 63(3–4), 153–179. <https://doi.org/10.1159/000503825>
- Van Maanen, J., Sørensen, J. B., & Mitchell, T. R. (2007). The interplay between theory and method. *Academy of Management Review*, 32(4), 1145–1154. <https://doi.org/10.5465/AMR.2007.26586080>
- Vogel, I. (2012). Review of the use of 'Theory of Change' in international development. Retrieved September 29, 2022, from https://www.theoryofchange.org/pdf/DFID_ToC_Review_VogelV7.pdf.
- Woolcock, M. (2009). Toward a plurality of methods in project evaluation: A contextualised approach to understanding impact trajectories and efficacy. *Journal of Development Effectiveness*, 1(1), 1–14. <https://doi.org/10.1080/19439340902727719>

Sanjeev Sridharan is Professor of Health Policy Evaluation at the University of Hawaii at Manoa. Previously he was the Country Lead, Learning Systems and Systems Evaluation at the India Country Office of the Bill and Melinda Gates Foundation. Prior to this position, Sanjeev was Director of the Evaluation Centre for Complex Health Interventions at St. Michaels Hospital and Associate Professor at the Department of Health Policy, Management and Evaluation at the University of Toronto. He is a former associate editor of the *American Journal of Evaluation* and has been on boards of the *Canadian Journal of Program Evaluation*, *New Directions for Evaluation* and *Evaluation and Program Planning*.

April Nakaima is a Research Officer at The Evaluation Centre for Complex Health Interventions at the University of Toronto in Canada. She has co-led evaluations of poverty reduction initiatives, brain health community initiatives, and dance activity health promotion. April has co-edited volumes in *Evaluation and Program Planning*, *New Directions for Evaluation*, and the *Journal of MultiDisciplinary Evaluation*.